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A Compact, Broadband, and Calculable Electromagnetic Field Source for Quality Assurance in EMC Testing

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***Abstract*—The aim of this article is to describe the design, realization, and characterization of a compact, broadband, and calculable electromagnetic field source. The source has been extensively used as a traveling reference radiator in proficiency tests of radiated emission measurements. The result of its characterization has been validated by more than 500 measurements over the frequency range from 50 to 6000 MHz in semianechoic (3 and 10 m) and fully anechoic (3 m) chambers. The use of the source as a reference radiator for emission measurements could be extended to other testing facilities, such as transverse electromagnetic waveguides and reverberation chambers.**

***Index Terms*—Anechoic chamber, comb generator, electromagnetic compatibility (EMC), measurement uncertainty, proficiency testing, quality assurance, radiated emissions, semianechoic chamber, traceability of measurement results.**

I. Introduction

COMPACT broadband reference sources of electromagnetic field are used for assuring the quality of the measurement results produced by testing laboratories. In the current practice of electromagnetic compatibility (EMC) testing laboratories, a stable, but uncalibrated, reference source is periodically measured to highlight minor and major variations in the operation of the measurement chain. This permits to quantify the nonrepeatability of the measurement chain and to identify possible defects or malfunctions. [1] is a pioneering and seminal work describing the electrical structure and use of such a stable and uncalibrated source.

The standard [2] requires that accredited testing laboratories implement means for ensuring the validity of test results. The successful participation in proficiency tests is the primary method in [2] for providing evidence that valid test results are produced. A proficiency test is an interlaboratory comparison designed to assess the competence of testing laboratories. The design and operation of a proficiency test are regulated by the standard [3]. Accredited EMC testing laboratories are not an exception, being subject to these same rules.

A significant amount of literature can be found dealing with interlaboratory comparisons and pioneering activity of proficiency testing in EMC emission measurements. A comprehensive analysis of this literature can be found in [4]. Reference values of the quantities (voltage and electric field) generated by the traveling sample are typically obtained at the end of the interlaboratory comparison as a (robust) average and standard deviation of the measurement results provided the participating laboratories. Limited in number but recent contributions extend the application of proficiency testing to conducted [5] and radiated [6] EMC immunity tests.

The idea of using calibrated sources of electromagnetic field as traveling samples in EMC radiated emission proficiency tests is for the first time introduced in [7]. One advantage is to allow participation of laboratories with different facilities (semianechoic and fully anechoic chambers) and testing at different distances (3 or 10 m). Calibrated (preassigned) reference values can be used together with calculated conversion values for the different testing environments and measurement conditions. Another advantage is the possibility of immediately providing the outcome of the proficiency test to those laboratories having this need, instead of waiting the end of the activity, whose duration is about six months.

In this contribution, the structure of a compact, broadband, and calculable electromagnetic field source that has been designed and manufactured by the authors and extensively used for proficiency tests of EMC radiated emission measurements is described. Mechanical, electromagnetic, and electronic aspects of the design and manufacturing are dealt with. We believe that disclosing these details can be of inspiration for those having practical interest in promoting quality assurance in EMC testing. Preliminary proficiency test results obtained by using the field source described here were reported in [4], where the focus is mainly on the methodological aspects of the operation of proficiency tests based on the use of traveling samples with preassigned reference values.

An overview of the process that was followed to obtain the electromagnetic field radiated by the source in the presence (semianechoic chamber) or absence (fully anechoic chamber) of a large ground plane is presented. Furthermore, the main steps of the quantification of the uncertainty on that field are reported.

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Results are presented for the two most common measurement distances of 3 and 10 m. The calibrated reference values in the presence of a ground plane are obtained assuming height scanning of the receiving antenna from 1 to 4 m for maximum detection. Such measurement technique is the one required by the worldwide adopted standards issued by the Comité International Spécial des Radioélectriques (CISPR), and, in particular, by the basic standard [8].

Finally, calibrated reference values are compared with the results of several proficiency tests carried out by using the reference source as a traveling sample. More than 500 measurements spanning the full frequency range covered by the reference source were gathered and processed by using standard statistical methods [9].

The information on the characterization of the source and the confirmation of its validity through the comparison with the results of the proficiency tests is given here for the purpose of highlighting the value of this piece of hardware in the context of EMC metrology and quality assurance. We do not enter here into the details of the characterization process and into quantification of the uncertainty of the electromagnetic field. A full description of these aspects is provided in [10] (simulation and measurement in the far field) and [11] (simulation and measurement in the near field).

This article suggests original and reliable methods for characterizing appropriately designed reference sources of electromagnetic field. Such methods do not require the use of a "golden site," but the coordinated use of an electromagnetic field simulator and near- and far-field measurements. A significant result is that the uncertainty on the electromagnetic field generated by the reference source is about half the measuring instrumentation uncertainty reported in [12] for far-field measurements.

The rest of this article is organized as follows. In Section II, the general design criteria and the desired and obtained characteristics of the reference source are described. The reference source is the result of the integration of a comb generator and a biconical antenna. In Section III, the antenna and its parameters are, respectively, described and quantified. Section IV outlines the comb generator, its principle of operation, and its circuit schematic. In Section V, the numerical and experimental process followed to quantify the uncertainty on the radiated field is described. Section VI details the use of the reference source for proficiency testing. The measurement results produced by the participants in the proficiency tests are compared with the predicted electromagnetic field, in order to demonstrate the validity of the characterization process. Finally Section VII concludes this article.

II. Design Criteria and Characteristics

The reference source is the combination of a comb generator and an antenna. Both the generator and antenna were designed in order to meet given requirements, which are presented in the following.

A broadband operation is necessary in order to cover the frequency range of radiated emission measurements with a single reference source. Since most of the EMC testing laboratories in Europe are accredited to IEC-CISPR standards requiring radiated emission measurements up to 6000 MHz; this is the desired, obtained, and maximum frequency of operation of the reference source.

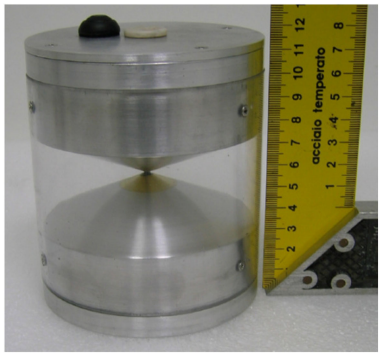


Fig. 1. Reference source.

In order to eliminate cables, which are a major source of nonreproducibility in measurement of radiated emission in the frequency range up to 300 MHz (mainly due to residual antenna imbalance, see [13]), the comb generator feeding the antenna is integrated in the same antenna. A further positive effect of this design choice is that the antenna balun, which is a critical component, is not necessary. The generator is battery operated and generates a comb of frequencies spaced by 50 MHz. The operating time, without recharging, is 4 h.

The electromagnetic field strength generated by the reference source (at 3 and 10 m) has to be comparable with the radiated emission limits (e.g., those set in [14] for information technology equipment). The field strength radiated by the reference source is comprised between about 26 dB (μV/m) (at the minimum frequency of 50 MHz and maximum distance of 10 m) and 87 dB (μV/m) (at the resonant frequency of the reference source and 3 m distance).

Transportation of the reference source from one laboratory to another must be safe (minimum mechanical stress) and inexpensive. This requires a compact and lightweight design. The reference source occupies the volume of a cylinder whose base diameter is 10 cm and its height is 12 cm. Its weight is 0.8 kg.

The dominant contribution to measurement uncertainty of radiated emission is the nonideality of the test facility due to reflection from the walls (see [12]). In order to uniformly illuminate the walls of the chamber, the reference source has axial symmetry and uniform pattern on the H-plane. The E-plane pattern must be smooth, with a dominant lobe in the whole frequency range in order to facilitate the characterization of the reference source and prediction of the radiated electromagnetic field.

Fig. 1 shows a picture of the reference source. The switch of the internal comb generator (black) and the universal serial bus (USB) connector for the battery charger (white) are visible on the top of the sample. A picture of the interior of the reference source is shown in Fig. 2. The comb generator and the lithium battery are hosted in the antenna (left side in Fig. 2) and closed by a cap at the top (right side of Fig. 2).

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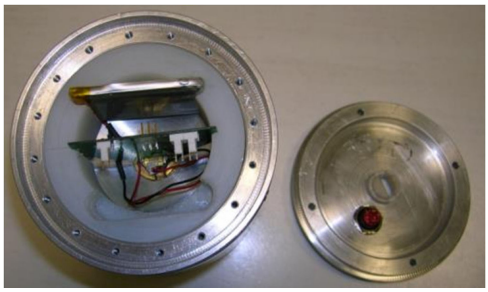


Fig. 2. Interior of the cone hosting the comb generator and the battery (left) and the cap of the cone (right).

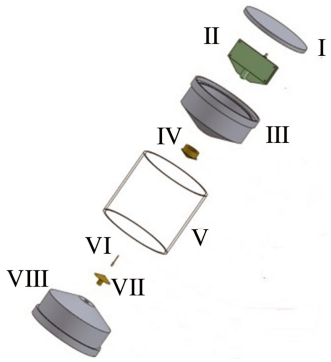


Fig. 3. Exploded view of the reference source. I—Cap; II—Comb generator; III—Upper cone; IV—Brass adapter (upper cone); V—Polycarbonate holder; VI—Pin; VII—Brass adapter (lower cone); and VIII—Lower cone.

The exploded view of the reference source is shown in Fig. 3, where the various parts are highlighted. The cones of the biconical antenna (III and VIII) are in aluminum in order to achieve a lightweight structure. The hot terminal of the comb generator (II) is soldered through a pin (VI) and a brass adapter (VII) to the lower cone (VIII). The cold terminal of the comb generator is soldered to the upper cone (III) through another brass adapter (IV). The upper cone is closed by a cap (I). The whole structure is framed within a polycarbonate (low relative dielectric constant $\varepsilon_r = 3.1$, low loss factor $\tan\delta = 0.01$) holder (V).

III. Antenna

The shape of the antenna that best matched the requirements listed above is the biconical one. The angle of the cone is $\psi = 134°$. This is the angle that corresponds to a characteristic impedance $Z_0 = 50\ \Omega$ for the transverse electromagnetic mode of a biconical transmission line (TL), where

$$Z_0 = \frac{1}{\pi}\sqrt{\frac{\mu_0}{\varepsilon_0}}\ln\cot\frac{\psi}{4}.$$

Here, μ_0 and ε_0 are the magnetic permeability and electrical permittivity of free space, respectively. This angle demonstrated to be the one for which the best match could be achieved, according to numerical simulations, for frequencies below 6000 MHz and for a base diameter of 10 cm. The upper cone has a cylindrical extension whose height is 3.4 cm (see Fig. 1). This extension is necessary in order to provide room for the comb generator and battery inside the same cone. The other cone has been equally extended in order to preserve symmetry of the E-plane pattern. These extensions improved matching at low frequency and had a negligible effect on matching above 2 GHz. Fig. 4 shows the calculated reflection coefficient of the antenna input (expressed as return loss, in dB). Calculations were carried out by using a commercial implementation of the method of moments (see Sections V-B and V-C for details). Note that the return loss is larger than 6 dB above 650 MHz, where the comb generator power starts to decrease with increasing frequency (see Fig. 6). The net result is that the radiated power is sufficient to produce the desired field strength level.

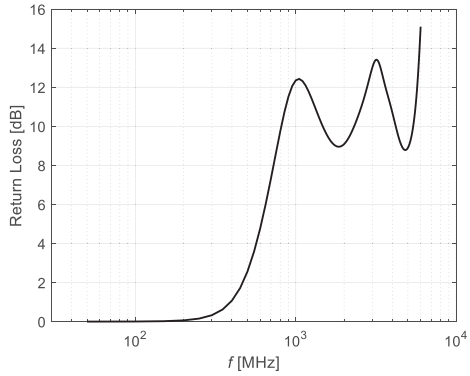


Fig. 4. Return loss of the antenna.

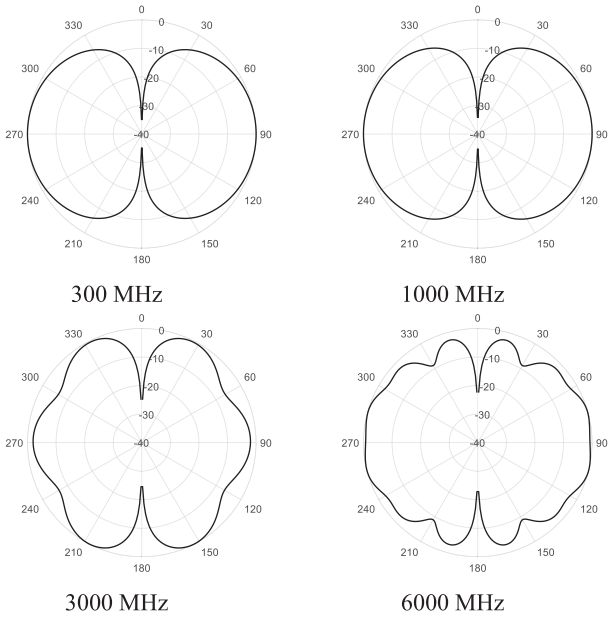


Fig. 5. Radiation pattern of the antenna at 300, 1000, 3000, and 6000 MHz.

The calculated E-plane radiation pattern is represented in Fig. 5 at four sample frequencies: 300 MHz (upper left plot), 1000 MHz (upper right plot), 3000 MHz (lower left plot), and 6000 MHz (lower right plot). The radiation pattern exhibits the typical short dipole $\sin\theta$ (θ is the pattern angle) behavior up to 1000 MHz. The radiation pattern is distorted but smooth and

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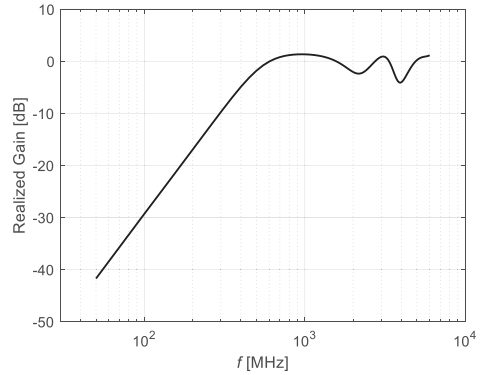


Fig. 6. Realized gain of the antenna.

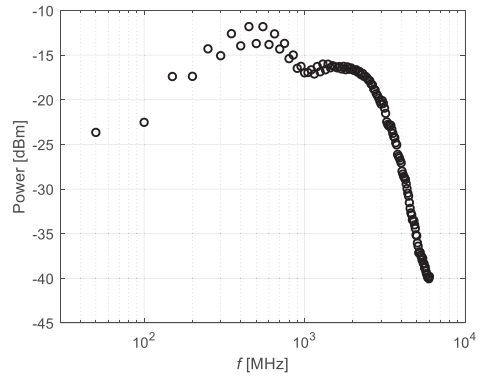


Fig. 7. Power that the comb generator delivers to a 50-Ω load. The harmonics spacing is 50 MHz.

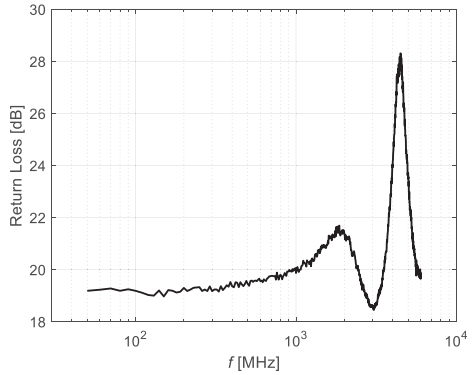


Fig. 8. Return loss of the output of the comb generator.

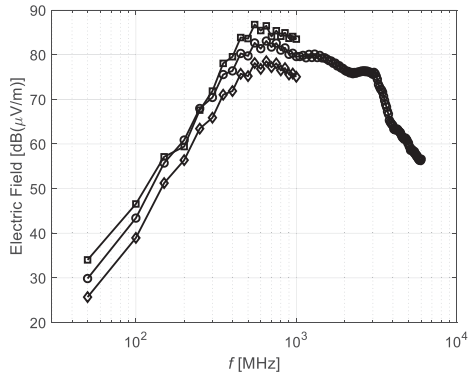


Fig. 9. Radiated electric field in a fully anechoic chamber (circles) at 3 m distance and in a semianechoic chamber at 3 m (squares) and 10 m (diamonds) distance.

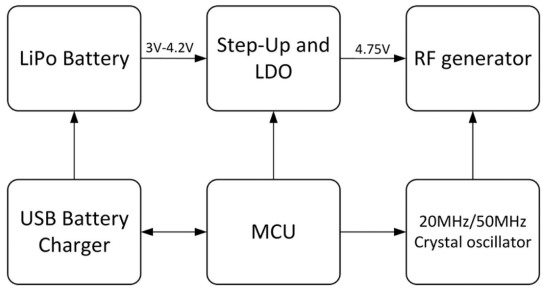


Fig. 10. Block diagram of the electronic circuit of the comb generator.

essentially split into two main lobes above 1000 MHz and up to 6000 MHz.

The realized gain in free space and in the direction $\theta = 90°$ is shown in Fig. 6. It is seen that the realized gain increases at 40 dB/dec for frequencies comprised between 50 and 500 MHz because the antenna behaves as a short dipole at these frequencies. The antenna is fairly matched above 500 MHz (see Fig. 4), and the realized gain is about 0 dB.

The power that the comb generator delivers to a 50-Ω load is represented in Fig. 7 at each frequency multiple of 50 MHz, up to 6000 MHz. The power decreases above 3 GHz, from which we deduce that the duration of the generated impulse is about 100 ps.

The return loss of the output of the comb generator is shown in Fig. 8, as measured with a vector network analyzer. The return loss is larger than about 19 dB, thus producing a relatively low mismatch error [15]. Finally, the electric field radiated in vertical polarization at 3 and 10 m distance in a semianechoic chamber (i.e., in an ideal empty space over an infinite and perfect conducting ground plane) and at 3 m distance in a fully anechoic chamber (i.e., in free space) are shown in Fig. 9. The magnitude of the vertical component of the electric field, as measured by a linearly polarized antenna, was calculated and represented in Fig. 9. Calculation for the semianechoic chamber takes height scanning from 1 to 4 m for maximum detection into account. Only the fully anechoic chamber is admitted for radiated emission measurements above 1000 MHz [8]. Hence, the prediction of the electric field stops at 1 GHz for the semianechoic chamber, whereas it reaches 6000 MHz for the fully anechoic chamber.

IV. Comb Generator

The general block diagram of the electronic circuit designed for this generator is depicted in Fig. 10.

The circuit takes the power from a lithium polymer (LiPo) battery. A step-up dc–dc converter stabilizes the variable cell voltage (3–4.2 V) to a constant 5.1-V board supply. Given the switching topology of the converter, a residual ripple is expected

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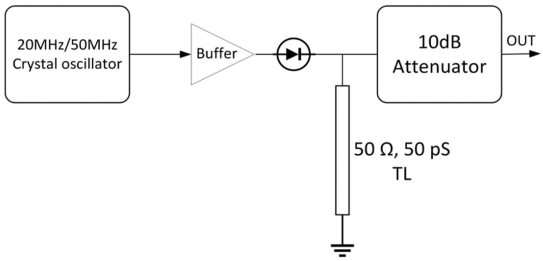


Fig. 11. RF generator circuit.

on this supply, which is thus not suitable to feed the analog radio frequency (RF) generator. The analog power rail is derived from the 5.1-V rail through the use of a low drop-out linear regulator, which generates a constant and low-noise 4.75-V supply rail. A LiPo battery charger circuit is implemented on board. The battery can be charged through a micro-USB connector, exposed on the flat top side of the antenna, by means of a standard cell-phone charger. An ultralow power microcontroller unit (MCU) is in charge of supervising both the power supply and the RF generator.

The MCU continuously measures the voltage of the battery, thus verifying the charge status. Whenever the battery voltage drops below a safety level, which guarantees a correct board-voltage regulation, the MCU shuts down the RF generator and goes into a low-power sleep state, waiting for the battery to be charged. This controlled shut down allows us to perform RF output measurements if, and only if, the generator power supply is known to be stable, which guarantees that the output amplitude is into the specified working range.

The RF generator is fed by two different crystal oscillators, at frequencies of 20 and 50 MHz. Only the 50 MHz frequency was employed in the applications described here. The active source is selected by the MCU: the user can select which frequency is to be used by pressing the black push button on the flat top of the antenna. The same button is used to turn ON and shut OFF the generator. One blinking LED, integrated in the button, signals the current operating modality of the system (OFF, 20-MHz line spacing, 50-MHz line spacing) by varying the blinking pattern.

The RF generator circuit (see Fig. 11) is based on a step-recovery diode (SRD) (GC2510-00, Microsemi Corp., Aliso Viejo, CA, USA) coupled to a short-circuited TL, whose length, combined with the transition time of the diode, defines the pulse duration. A square wave is generated, whose frequency is based on the desired harmonics spacing, and a buffer stage, composed by four SN74LVC2G32 chips, drives the anode of the SRD. The combined current capability of the buffers is more than 200 mA. During the high level of the square wave, the forward-biased SRD injects its current into the transverse short-circuited TL. When the square wave reverses its polarity, the current into the SRD and the TL reverses too, until all the charge is removed from the SRD junction. At this point, the SRD abruptly stops conducting, and the current in the TL deviates toward the output, generating an impulse having positive polarity. The duration of this impulse is approximately the sum of two times the propagation delay of the TL and the transition time of the diode. A 10-dB attenuator is added between the TL and the SMA connector in order to effectively damp multiple reflections between the antenna input and TL output and to provide the generator output match.

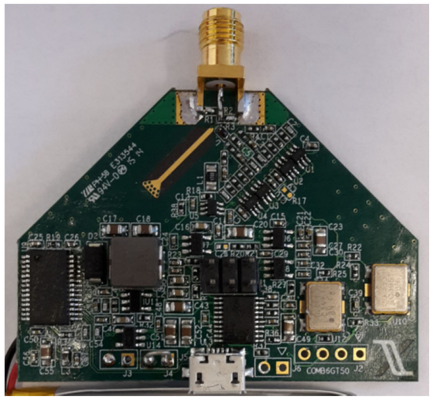


Fig. 12. PCB of the comb generator and components' side.

The circuit has been realized on a four-layer printed circuit board (PCB) (see Fig. 12), using a high-frequency low-loss dielectric material (RT5880, Rogers Corp., Chandler, AZ, USA). All the components are surface mount, to minimize parasitic parameters that would corrupt the integrity of the pulse waveform. The RF passive components are of 0402 size (1×0.5 mm). The TL is designed with an impedance of 50 Ω and a propagation delay of 50 ps (1-cm length). The end of the line is tapered to a via array for effective and smooth ground connection. The TL is not covered by the solder mask, to prevent nonidealities introduced by this dielectric material, whose thickness and dielectric properties are neither specified nor controllable during the manufacturing process.

V. Uncertainty on the Radiated Field

The electric field radiated by the reference source is known with an amount of uncertainty. Such uncertainty must be quantified in order to make the electric field generated by the reference source traceable to known standards. We here provide a summary of the process of uncertainty quantification to make this document self-supporting. Details can be found in [10] and [11].

A. Uncertainty on the Power

The power P that the comb generator delivers to a 50-Ω load was calibrated through the substitution method comparing, by using a spectrum analyzer, the amplitude of each harmonic with the amplitude of a sinewave generated by a calibrated signal generator. The standard uncertainty of the calibrated power is $u_C = 0.2$ dB (at 20 °C). This contribution to measurement uncertainty neglects the short-term instability (within 2 h from switch-ON) of the comb generator, which has been evaluated as $u_S = 0.12$ dB (standard uncertainty). An additional contribution to uncertainty on power comes from instability due to temperature fluctuations within the range comprised between 15 and 25 °C, which has been evaluated as $u_T = 0.29$ dB. The

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Fig. 13. Measurement of the insertion loss between the biconical antenna terminals and the output of a highly symmetric loop antenna.

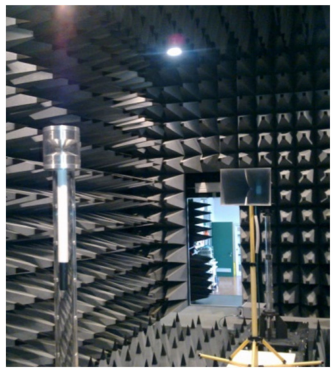


Fig. 14. Measurement of the electric field generated by the biconical antenna through calibrated receiving antennas in an anechoic chamber.

combined standard uncertainty on power is therefore

$$u_P = \sqrt{u_C^2 + u_S^2 + u_T^2} = 0.4 \text{ dB}. \quad (1)$$

B. Uncertainty on the Electromagnetic Model of the Antenna (50–300 MHz)

The computer aided design electromagnetic model (briefly, electromagnetic model) of the biconical antenna that has been submitted to numerical simulations (by using FEKO [16]) has been validated through comparison between predictions and measurements. The uncertainty on the electromagnetic model of the biconical antenna between 50 and 300 MHz has been calculated by comparing the measured and calculated insertion loss between the biconical antenna input (i.e., the comb generator output) and the output of a highly symmetric shielded loop antenna specifically designed and realized for this purpose (see Fig. 13).

The loop is a square whose side is 7 cm and it is made of a single piece of semirigid 50-Ω coaxial cable (type RG 402). The external diameter of the coaxial cable is 3.6 mm. The gap in the shield [17] is in the middle of the side opposite to the neck. The structure is highly symmetric to minimize common mode current effects [18]. The two ports of the loop antenna, corresponding to the two extremes of the coaxial cable, are both loaded by 50 Ω: one 50 Ω is the input resistance of the receiver; and the other 50 Ω is a dummy load. The side of the loop antenna opposite to the neck touches the polycarbonate holder of the biconical antenna during insertion loss measurement and the plane of the loop antenna contains the axis of the biconical antenna (see Fig. 10). A variable step attenuator is inserted in the measurement chain in order to avoid changing the vertical scale settings (reference level and decibels per division) of the spectrum analyzer used for power measurement, which would introduce switching inaccuracies.

The standard uncertainty of the insertion loss measurement was evaluated as 0.7 dB. The root-mean-square deviation between the measured and predicted insertion losses was 0.15 dB at the six frequencies comprised between 50 and 300 MHz, i.e., negligible with respect to the standard uncertainty on the insertion loss measurement. The standard uncertainty of the electromagnetic model of the antenna in the frequency range between 50 and 300 MHz was therefore quantified as $u_{\text{EM}} = 0.7$ dB.

TABLE I
RECEIVING ANTENNAS, ROOT-MEAN-SQUARE DEVIATION BETWEEN PREDICTIONS AND MEASUREMENTS, AND UNCERTAINTY OF THE ELECTROMAGNETIC MODEL OF THE BICONICAL ANTENNA IN THE FREQUENCY INTERVALS INVESTIGATED THROUGH MEASUREMENTS IN THE ANECHOIC CHAMBER

Frequency (MHz)	Receiving antennas	RMS deviation (dB)	u_{EM} (dB)
350-1000	A, B, C	0.70	1.7
1050-3000	B, C, D	0.56	1.2
3050-6000	C, D	0.61	1.3

C. Uncertainty on the Electromagnetic Model (350–6000 MHz)

A similar procedure was followed to the one described in Section V-B in order to determine the standard uncertainty of the electromagnetic model of the biconical antenna in the frequency range between 350 and 6000 MHz.

In this frequency range, the electric field measured with calibrated receiving antennas in a fully anechoic chamber (see Fig. 14) is compared with the numerical prediction of the field in free space. Four calibrated receiving antennas were used in order to cover the full frequency range. The frequency ranges covered by the different antennas overlapped, therefore at least two measurement results were available for each harmonic frequency. Averaging these measurement results improves measurement accuracy.

The four selected receiving antennas were (the frequency range covered by each antenna and the expanded calibration uncertainty U at 95% coverage probability are indicated within brackets): A—Biconical (150–1000 MHz and $U = 2$ dB); B—Log periodic (200–3000 MHz and $U = 0.7$ dB); C—Horn 1 (500–6000 MHz and $U = 2$ dB); and D—Horn 2 (1–18 GHz and $U = 2$ dB).

The combination of the receiving antennas used for the three frequency intervals covering the full span from 350 to 6000 MHz is detailed in Table I. The root-mean-square deviation between prediction and measurements and the value of the standard

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TABLE II
CONTRIBUTIONS TO THE UNCERTAINTY ON THE ELECTRIC FIELD RADIATED BY THE BICONICAL ANTENNA AND COMBINED STANDARD UNCERTAINTY

Frequency (MHz)	u_P (dB)	u_M (dB)	u_{EM} (dB)	u_E (dB)
50-300	0.4	0.7	0.7	1.1
350-1000	0.4	0.7	1.7	1.9
1050-3000	0.4	0.3	1.2	1.3
3050-6000	0.4	0.3	1.3	1.4

uncertainty of the electromagnetic model are also reported in Table I.

D. Uncertainty on the Radiated Field (50–6000 MHz)

The uncertainty on the field radiated by the reference source was obtained by using the following measurement model equation:

$$E = P + 107 + M + \delta E \tag{2}$$

where E is the amplitude of the radiated field [in dB(μV/m)], P (in dB·m) is the power that the comb generator delivers to a 50-Ω load, M (in dB) is the correction for mismatch between the comb generator output and the biconical antenna input, and δE [in dB(1/m)] is a correction numerically computed by using the electromagnetic model of the antenna. The expected value of the mismatch correction is 0 dB [15]. The standard uncertainty of the mismatch correction (u_M, in dB) was calculated assuming a return loss of the comb generator output of 19 dB (see Fig. 8) and return losses of the input of the biconical antenna of 0 and 8 dB in the frequency ranges from 50 to 1000 MHz and from 1000 to 6000 MHz, respectively (see Fig. 4). The computed correction δE depends on frequency, test facility (fully or semianechoic chamber), measurement distance (3 or 10 m), measurement method (height scanning in a semianechoic chamber and fixed height in a fully anechoic chamber), and input mismatch of the biconical antenna. The standard uncertainty on δE is the standard uncertainty on the electromagnetic model u_{EM}.

The combined standard uncertainty on the radiated field is calculated as

$$u_E = \sqrt{u_P^2 + u_M^2 + u_{\mathrm{EM}}^2}. \tag{3}$$

The values of the three contributions u_P, u_M, and $u_{\mathrm{EM}}$are summarized in Table II along with the corresponding combined standard uncertainty u_E calculated by using (3), in each frequency range.

Since the combined standard uncertainty in the frequency range between 1050 and 3000 MHz (1.3 dB) is nearly equal to that in the frequency range from 3050 to 6000 MHz (1.4 dB), the larger of the two values is taken as the representative one of the full range from 1000 to 6000 MHz.

VI. USE OF THE REFERENCE SOURCE IN PROFICIENCY TESTING

The reference source was used in interlaboratory comparisons of radiated emission measurements for the assessment of the competence of EMC testing laboratories (proficiency tests). The use of the reference source for this purpose is already documented in [7]. The measurement results obtained during four rounds of proficiency tests carried out in 2015, 2016, 2017, and 2018 are here compared with the numerically computed reference values of the electric field and the corresponding uncertainty.

TABLE III
NUMBER OF PARTICIPANTS IN FOUR PROFICIENCY TESTS, THE GENERATED NUMBER OF MEASUREMENT RESULTS, AND THE TYPE OF TEST FACILITIES (SEMIANECHOIC CHAMBER—SAC; FULLY ANECHOIC CHAMBER—FAC)

Year	Participants	Results	SAC 3 m	SAC 10 m	FAC 3 m
2015	9	86	6	1	2
2016	13	118	8	2	3
2017	11	102	7	0	4
2018	24	224	17	2	5

The number of participating laboratories in each proficiency test and the type of test facility is detailed in Table III. In total, ten frequencies were investigated in each proficiency test, covering the full range up to 6000 MHz. The total number of measurement results provided by the participants is 530 because some of the 57 participants could not perform measurements up to 6 GHz, thus providing less than ten measurement results.

Only two frequencies of the first proficiency test in 2015 were reused in 2016 and 2017, namely 50 and 200 MHz. The proficiency test carried out in 2018 reused the same frequencies as those of the 2015 proficiency test for the purpose of comparing measurement results obtained after three years. Hence, the number of distinct investigated frequencies is 28. One laboratory participated in the four proficiency tests, six laboratories participated in three of the four proficiency tests, and seven laboratories participated in two of the four proficiency tests. Among the 57 participants, 34 are independent test houses (different testing facilities, instrumentation, and personnel). Some of the laboratories that participated in multiple editions used different testing facilities, instrumentation, and personnel. In any case, correlations cannot be excluded among the sets of measured values corresponding to each proficiency test.

The measurement result provided by each participant is compared with the reference value corresponding to the test facility and distance (SAC 3 m, SAC 10 m, or FAC 3 m) employed by the same participant. The difference between the measurement result provided by the ith participant and the reference value is δ_i. The δ_i values are statistically processed by using the robust (outlier resistant) statistical procedure named "Algorithm A" in [9], Annex C in order to determine a robust average δ^* and a robust standard deviation s^*. Ideally $\delta^* = s^* = 0$ dB. The robust standard deviation of the robust average is given by $1.25s^*/\sqrt{p}$ [9], where p is the number of participants ($i = 1, 2, \ldots, p$, note that for some frequencies, the number of participants is less that the value that appears in the second column of Table III because, as mentioned before, not all the participants were able to cover the entire frequency range).

The agreement between the predicted and measured electric field has been determined for the whole set of participants'

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TABLE IV
VALUES OF THE STATISTIC Δ FOR THE FOUR PROFICIENCY TESTS AND EACH INVESTIGATED FREQUENCY

2015		2016		2017		2018	
f (MHz)	Δ	f (MHz)	Δ	f (MHz)	Δ	f (MHz)	Δ
50	-0.1	100	1.5	50	0.1	50	0.4
150	0.3	200	0.3	200	0.2	150	0.6
250	0.9	300	-0.3	400	0.1	250	0.2
500	0.4	600	0.7	700	-0.1	500	-1.2
750	0.7	800	0.7	900	0.1	750	-0.2
950	1.1	1100	0.1	1800	-0.3	950	0.4
2000	0.1	2200	1.4	2450	-0.2	2000	-0.4
3000	0.4	3500	0.8	3300	-1.0	3000	0.1
5000	0.7	5200	0.7	4300	-0.1	5000	-0.9
6000	0.7	5800	0.4	5900	-0.4	6000	-0.9

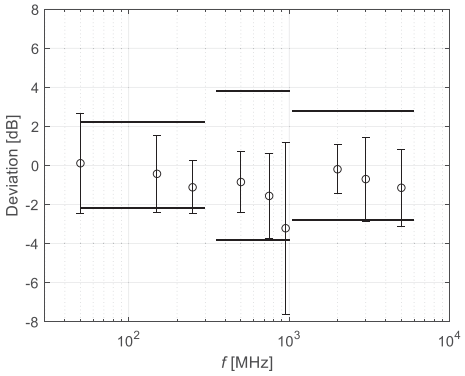


Fig. 15. Deviation between measured and predicted values of the electric field, as a function of frequency. Circles correspond to δ^*, and the half-length of the error bar is $2(1.25s^*/\sqrt{p})$. The zero-level reference corresponds to the predicted value and the horizontal segments to $\pm 2u_E$. 2015 proficiency test.

values based on a performance statistic given by

$$\Delta = \frac{\delta^*}{\sqrt{\left(1.25s^*/\sqrt{p}\right)^2 + u_E^2}}. \tag{4}$$

The work in [9, clause 7.8 and Annex E.7] provides interpretation to this statistic in terms of "no anomaly," "warning signal," and "action signal." If $|\Delta| < 2$, we have no anomaly (agreement). If $|\Delta| > 2$, we have an action signal (disagreement). If $2 < |\Delta| < 3$, we have a warning signal (nothing can be concluded about agreement or disagreement). The values of the statistic Δ for four proficiency tests at each investigated frequency are reported in Table IV. It is seen that the maximum value of $|\Delta|$ is 1.5, hence no anomaly is detected. In other words, the predicted and measured values of the electric field strength agree with each other.

A visual representation of the comparison between predicted and measured values of the electric field strength leading to the same conclusion can be obtained from Figs. 15–18 (each figure corresponds to one of the four proficiency tests). The robust average δ^* of the deviation from the reference value is represented in each figure (with circles, squares, diamonds, and asterisks). The half-length of the error bar corresponds to two times $1.25s^*/\sqrt{p}$. The horizontal segments correspond to plus/minus the expanded uncertainty (at 95% coverage probability, assuming normal distribution) of the reference value (i.e., $\pm 2u_E$, see Table II).

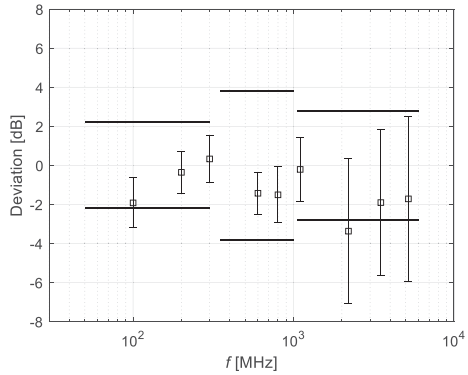


Fig. 16. Same as Fig. 15 but for the 2016 proficiency test.

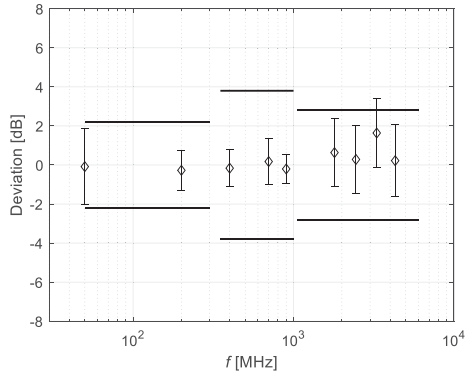


Fig. 17. Same as Fig. 15 but for the 2017 proficiency test.

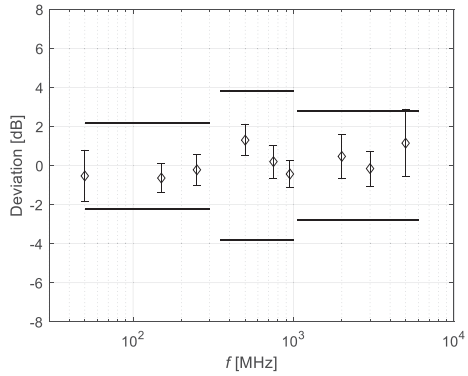


Fig. 18. Same as Fig. 15 but for the 2018 proficiency test.

VII. CONCLUSION

In this article, a compact and broadband reference source of electromagnetic field that generates a known field strength level is a quite useful device that can be used for periodical and rapid verification of the measurement chain of a testing laboratory. Since the absolute value of the field is known, the verification does not require that the measurement chain nominally remains

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the same at each periodical verification. The results obtained by using the reference source as a traveling sample for proficiency testing confirmed the validity of the electromagnetic field predictions and, above all, the usefulness and effectiveness of the device for such measurement exercises. Another interesting application is the use of the reference source for investigating the traceability of measurements in transverse electromagnetic waveguides and reverberation chambers through interlaboratory comparisons having preassigned reference values of radiated electromagnetic field and radiated power.

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